Product 1: TOU-Aware Energy Management and Dispatch

Oct 30, 2026 · @Onyxleaf Claude AI and Mechanical

Summary

Verdict: pursue Product 1 as a field pilot, reframed from a "TOU optimiser" to a cost-aware dispatch optimiser for sites with outages and gensets. Pure TOU arbitrage in Pakistan nets Rs 4.8 to 10.3 per kWh, about the Rs 7.5 to 10.8 of battery wear it causes.

A 100 kWh battery at a B-2 industrial site earns about Rs 1,031 a day from grid-charged TOU arbitrage (October 2026 inputs). The same battery earns about Rs 2,346 a day storing net-billed PV, and about Rs 11,685 on a day it replaces diesel. So the product must win outage, genset and self-consumption decisions; TOU timing is one input among four.

Enerlyticslab monitors these assets and forecasts load shedding, but no source shows it sending setpoints to inverters, batteries or gensets. NeoTAQ, sold by Green Fuel Energy, already runs rule-based hybrid and battery dispatch at a claimed 50+ Pakistani sites. No vendor-neutral, forecast-driven, tariff-aware optimiser was found in Pakistan or Saudi Arabia.

#	Finding	Key number	Source
1	TOU arbitrage barely covers battery wear	B-2 nets Rs 10.3/kWh, A-2(c) Rs 4.8/kWh; wear Rs 7.5 to 10.8/kWh (6,000-cycle assumption)	IESCO tariff guide, SRO 279(I)/2026, SolarCitizen, Oct 2026
2	Net billing makes self-consumption the main grid value	Exports valued at Rs 8.13/kWh in NEPRA's Apr to Jun 2026 adjustment vs import Rs 35.68 to 43.82; PV shift worth Rs 23 to 35/kWh	NEPRA SRO 1501(I)/2026
3	Genset displacement dominates on outage days	Diesel Rs 400.67/L (30 Oct 2026) gives Rs 110 to 142/kWh; outage-day value 11 times arbitrage	Daily Pakistan, Enpower

#	Finding	Key number	Source
4	Maximum-demand fixed charges are a hidden stream	Rs 1,250/kW/month; a 25 kW MDI cut is about Rs 375,000/yr where MDI sets the charge	IESCO tariff guide
5	Evidence that an optimiser beats the inverter's own mode is thin	Hardware test in 4 reproduced homes: rule, decision-tree and MPC costs within 0.6%; for hourly schedulers, MPC's edge shrinks 69% when re-scored at 1 minute	Ruddick et al. 2024, Pinter et al. 2025
6	Saudi Arabia has no business TOU	Industrial 20 halalas/kWh (23 from 1 Jan 2027), export 5 halalas; about SAR 4,300 to 5,200/yr per 100 kWh battery	MEED, Haala Energy

Where the money is, ranked by rupees per kWh of battery throughput:

1. Outage and genset displacement: Rs 110 to 142/kWh. A kWh held for a forecast outage is worth about Rs 106 more than spending it on the B-2 peak.
2. Net-billing self-consumption: Rs 23 to 35/kWh. The inverter's self-use mode already captures most of it; the increment lies in charge timing.
3. Maximum-demand (MDI) fixed charge: Rs 1,250/kW/month, only where MDI sits above the sanctioned-load floor.
4. Pure TOU arbitrage: Rs 4.8 to 10.3/kWh gross, near zero after wear, and zero on units under the flat incremental package.

Where the money is not: schools and hospitals on A-3 have no TOU option, and D-2(b) tubewells see a Rs 0.85 spread. Legacy net-metered sites still net exports at about Rs 25, and Saudi C&I tariffs are flat.

The evidence gap is the main risk. MPC beat rules mainly where peak charges applied: 13.9% vs 5.2% in one simulation (Pérez-Piñeiro et al. 2023). No vendor A/B against an inverter's native TOU mode was found, so the pilot must measure that increment first.

Refined problem statement

The meeting's idea, "reproduce Enerlytics with TOU", needs six corrections once 2026 tariffs, outages and net billing are applied. The result is a dispatch problem with four value streams, of which TOU is the smallest.

What the research changes

- TOU applies only to some categories: A-1 residential at 5 kW and above, A-2(c) commercial, B-1(b) to B-4 industrial, C bulk and D agriculture (IESCO). Schools and hospitals on A-3 pay a flat Rs 42.48/kWh.
- The Incremental Consumption Package (ICP) prices industrial and agricultural units above baseline at a flat Rs 22.98 until about December 2028. Arbitrage on those units is worth zero; NEPRA reviewed the package on 5 October 2026 (Express Tribune).
- Load shedding sits in the evening peak. The April 2026 relief window ran 5 pm to 1 am, so reserve and arbitrage compete for the same kWh (Profit).
- Net billing (9 February 2026) splits sites in two. New prosumers export at about Rs 8 to 11; legacy net-metered sites kept near unit-for-unit netting under a draft amendment.
- Saudi Arabia has no business TOU and credits PV exports at 5 halalas.
- Off-grid and remote sites have no tariff at all; their decision is when to run the genset.

Problem statement

C&I and remote sites in Pakistan run PV, lithium batteries, unreliable grid and gensets on fixed inverter rules, and lose money daily. Four decisions drive it: when and from what to charge, when to discharge, when to run the genset, and what reserve to hold. Product 1 makes these decisions every 15 minutes from forecasts and the actual tariff, and proves the saving against the inverter's own mode.

Scope

In scope	Out of scope
Grid-tied C&I on TOU (A-2(c), B-1(b), B-2, B-3) with PV, battery and optional genset	PV structure design (on hold)
Flat-tariff sites with outages and gensets (A-3 hospitals and schools, A-2(b) shops)	Homes (free OEM apps; SolarAssistant-level price ceiling)
Remote diesel-hybrid sites (farms, off-grid industry) as pilot 1b	Utility-scale, front-of-meter and wholesale trading

In scope	Out of scope
Tariff and regime model: TOU windows, ICP, net billing vs net metering, MDI	Device drivers, edge hardware, register maps (Product 2)
PV, load and outage forecasts	Battery and PV sizing studies
MILP rolling-horizon optimiser with genset, outage, MDI and wear terms	EV charging and HVAC load control
Independent battery-state check used by the optimiser	Saudi C&I arbitrage (no TOU)
Plan delivery: daily inverter TOU slots first, 15-minute setpoints via Product 2 later	Grid-tied tubewells on D-2(b) (flat tariff)
Savings verification against the inverter's own mode	Telecom tower fleets as a first market

First target buyer

Segment 1 is grid-tied industrial and commercial sites on TOU tariffs B-2, B-3 or A-2(c) with PV, 100+ kWh of battery and a genset. They sit on feeders with load shedding; typical buyers are textile, food, beverage and rice-processing plants.

- They stack all four streams: outage reserve, PV self-consumption, MDI and some TOU timing.
- Industrial storage is growing: its share of lithium imports rose from 3.2% (2025) to 13.9% (January to May 2026) (Pakistan Solar Association).
- Named buyers exist: Liberty Mills (30 MWh), Gadoon and Sapphire Textile (25 MWh each), Lucky Cement (22.7 MWh), Shabbir Tiles (20 MWh).
- They have energy managers, buy on ROI, and are served by Green Fuel Energy, Enercon and Enerlyticslab.
- They mix brands: Haideri Beverages runs 7 Huawei and 6 SMA inverters with 5 gensets (GFE blog).

Segment 1b is remote diesel-hybrid sites, where the manager's "off-grid, remote" demand sits. Telecom fleets come later, because operators and towercos buy integrated systems such as Huawei's (1,000 Jazz sites) and Reon SPARK (5,500 towers).

Market context

Pakistan's tariff gives industrial TOU sites a Rs 12.85/kWh spread, while a new prosumer's export credit sits Rs 27.55 below the B-2 peak. That gap, plus genset fuel at Rs 110 to 142/kWh, sets where an optimiser earns money.

Tariffs in force (base rates, before adjustments and taxes)

One national schedule, SRO 279(I)/2026 (February 2026), covers all ex-WAPDA DISCOs and K-Electric until end-December 2026 (IESCO tariff guide; NEPRA decision, 11 Feb 2026).

Category	Peak Rs/kWh	Off-peak Rs/kWh	Spread	Break-even round trip	Fixed charge
A-1 residential TOU (5 kW+)	46.85	34.53	12.32	74%	Rs 675/kW/M on 50% of sanctioned load or MDI
A-2(c) commercial TOU	43.82	35.15	8.67	80%	Rs 1,250/kW/M on 25% of sanctioned load or MDI
A-2(b) commercial flat (5 kW+)	39.76	39.76	0	none	Rs 1,250/kW/M
A-3 general services (schools, hospitals)	42.48	42.48	0	none	Rs 1,000; no TOU option
B-1(b) industrial TOU (to 25 kW)	35.74	25.48	10.26	71%	Rs 1,000 (Geo reports Rs 1,250)
B-2(b) industrial TOU (25 to 500 kW)	35.68	22.83	12.85	64%	Rs 1,250/kW/M (MDI basis not verified)
B-3 industrial (11/33 kV)	35.68	23.67	12.01	66%	Not legible in fetched pages
B-4 industrial (66/132 kV)	35.68	23.38	12.30	66%	Not legible
C-1(c) bulk TOU	46.31	37.54	8.77	81%	Not found
D-2(b) agriculture TOU	29.54	28.69	0.85	97%	Rs 2,000 minimum

The February 2026 cut fell mostly on off-peak rates: B-2 off-peak dropped from Rs 27.41 to Rs 22.83 (Geo News). Quarterly adjustments apply to all hours, such as +Rs 0.5194/kWh billed September to November 2026, so they leave the spread unchanged (NEPRA SRO 1501(I)/2026).

Peak windows last 4 evening hours and move with the season. Per an unofficial GEPCO guide (14 June 2026), peak runs 5 to 9 pm in winter, 6 to 10 pm in spring and autumn, 7 to 11 pm in summer (GEPCO guide). K-Electric bills 6:30 to 10:30 pm from April to October (PKRevenue, 30 Jul 2026); no official monthly notice was found.

The ICP charges a flat Rs 22.98/kWh on industrial and agricultural consumption above baseline, at all hours, from December 2025 for three years. Consumption ran 25% to 35% above baseline in early 2026, triggering a review hearing on 5 October 2026 (Express Tribune, 6 Oct 2026). A World Bank-backed industrial TOU scheme cited in the earlier review could not be confirmed.

Net billing

NEPRA notified the Prosumer Regulations 2026 on 9 February 2026, valuing imports and exports separately (Dawn, 10 Feb 2026). New prosumers' exports earned about Rs 10 to 11 at launch. NEPRA's April to June 2026 quarterly adjustment accounted for net-billing exports at Rs 8.13/kWh and net-metering units at Rs 25.32, so both regimes run side by side (NEPRA SRO 1501(I)/2026).

- New contracts last 5 years; capacity is capped at sanctioned load; DISCOs stop at 80% transformer loading (Dawn).
- A 16 February draft keeps existing agreements on old terms until expiry; no source confirms it was finalised (Profit).
- One trade report gave the new export rate as Rs 13 (pv magazine); NEPRA's Rs 8.13 is used here.
- Headlines (pages not read) report proposals for TOU net billing and Rs 22/unit night exports (ProPakistani, 2 Aug 2026).

Load shedding

From 14 April 2026, a fuel-driven "Peak Relief Strategy" cut supply up to 2.25 hours a day between 5 pm and 1 am (Profit). HESCO and K-Electric were exempt, and it ended about 1 May 2026 when spot LNG arrived (The Nation).

"Economic" load shedding on high-loss feeders continues as anti-theft policy (ProPakistani, 30 Apr 2026). In April 2026, rural MEPCO areas saw 12 to 16 hours a day and Lahore 3 to 4 hours (Dawn, 13 Apr 2026). Telecom operators report up to 12 hours a day, with batteries lasting 1 to 2 hours before gensets start (PhoneWorld, 1 Jun 2026).

Fuel and genset cost

Diesel costs Rs 400.67/L from 30 October 2026, after several October revisions (Daily Pakistan, 29 Oct 2026). At 0.25 to 0.33 L/kWh, fuel costs Rs 100 to 132/kWh, plus Rs 5 to 15/kWh maintenance (Enpower, 9 May 2026). No captive-gas tariff for October 2026 was found.

Battery market

Lithium imports reached 4.6 GWh in 2025, up 220%, with 7.6 GWh cumulative (ESS News, 15 Jul 2026). Customs data give a lower 3.31 GWh for 2025, plus 2.08 GWh for January to May 2026 (Pakistan Solar Association). Cumulative stock to end-2025 splits residential 3.1 GWh, industrial 2.0, telecom 1.4, commercial 1.0.

Installed retail LFP costs Rs 45,000 to 65,000/kWh, typically Rs 52,000 (16 October 2026); no C&I system price was found (SolarCitizen). IEEFA modelled 4 to 6-year C&I paybacks (IEEFA, 2025). Off-grid solar exceeds 13,000 MW (Dawn, Feb 2026).

Telecom towers

Pakistan has over 55,000 towers, 9,200+ diesel thefts in 11 months, and nearly 20,000 towers hit by fuel-related outages (PhoneWorld, 1 Jun 2026). Huawei and Jazz finished solar, storage and energy management at 1,000 sites (13 MW) in April 2026 (DCD). A Rs 3bn green sukuk now funds tower solar and lithium storage (Profit, 21 Apr 2026).

Worked value example

Assumptions are the author's, not sourced: 100 kWh usable, one cycle a day, 90% round trip, at least 25 kW of load through the peak. Inputs: B-2 35.68/22.83, A-2(c) 43.82/35.15, export Rs 11 (Rs 8.13 sensitivity), diesel at 0.33 L/kWh plus Rs 10/kWh maintenance.

Stream	Arithmetic	Rs per day	Rs per year
TOU arbitrage, B-2	100 × 35.68 − 111.1 × 22.83	1,031	376,400 (365 days)
TOU arbitrage, A-2(c)	100 × 43.82 − 111.1 × 35.15	476	173,900
PV shift into B-2 peak	3,568 − 111.1 × 11	2,346 (2,665 at Rs 8.13)	703,700 (300 days)
PV shift into A-2(c) peak	4,382 − 1,222	3,160 (3,479 at Rs 8.13)	947,900
Genset displacement, outage day	100 × 142.2 − 2,536.7 recharge	11,685	per outage day
MDI cut of 25 kW	25 × 1,250 × 12	none	375,000 (if MDI binds)

Stacked at a B-2 site, 60 outage days, 245 PV-surplus days and 60 cloudy days give about Rs 1.34M a year. Against Rs 5.2M capex (Rs 52,000/kWh), that is a 3.9-year payback; without outages Rs 0.78M (6.7 years), and arbitrage alone Rs 0.38M (13.8 years). Over an assumed 6,000 cycles, wear costs Rs 7.5 to 10.8/kWh, so dispatch should cycle mainly for PV shift and outages.

Which part is the optimiser's increment

These figures value the battery, not the optimiser. An inverter in self-use mode already earns most of the PV-shift stream, so Product 1 earns on forecast-dependent decisions (author's arithmetic from the inputs above):

- Holding reserve for a forecast outage: about Rs 106/kWh more than discharging into the peak (142.2 − 35.68).
- Skipping overnight grid charging before a sunny day: avoids paying Rs 22.83 for energy that pushes PV out at Rs 8.13, about Rs 14.7/kWh.
- Grid pre-charging only on cloudy days: up to Rs 1,031 per 100 kWh at B-2.
- Avoiding low-load genset hours: HOMER's example curve burns about 0.41 L/kWh at 25% load and 0.31 at full load, about Rs 40/kWh (HOMER docs).
- Capping monthly MDI, and skipping cycles whose margin is below wear.

Saudi Arabia

Saudi Energy's tariffs are tiered and flat, with no TOU element found for C&I. From 1 January 2027, industrial rises from 20 to 23 halalas/kWh and commercial from 22/32 to 25/35 (MEED, 26 Oct 2026). Small-scale PV up to 2 MW earns 5 halalas for surplus, and off-grid systems need no licence (Haala Energy, 7 Oct 2026).

A SERA draft (October 2026) would allow self-consumption up to 50 MW, with 5 to 7 halala export credits and storage (SMM). A 100 kWh battery shifting PV earns about SAR 4,300 to 5,200 a year at industrial rates (author's arithmetic). Saudi demand therefore lies at remote diesel-hybrid sites and in export-cap control, not tariff arbitrage.

Enerlyticslab deep dive

Enerlyticslab is a Karachi cloud monitoring and analytics platform with about 3 to 10 staff, 10 asset modules and no published closed-loop control. Its rendered website, read in a browser on 31 October 2026, is the most reliable source (enerlyticslab.com).

Company facts

Item	Detail	Source
Legal name	Enerlytics Tech Lab (Pvt) Ltd	Google Play
Address	Suite 509, Fortune Center, Shahrah-e-Faisal (PECHS Block 6), Karachi; contact@enerlyticslab.com	Home page
Founded and size	2020; 3 or 10 employees (sources conflict)	RocketReach
Positioning	"Enabling Energy Intelligence Through Digitalization"; "a group of experienced energy consultants"	About
Target customers	Industrial and commercial facilities, renewable companies, ESCOs, asset managers	About
Traction	Android app 10+ downloads (updated 28 Sep 2026); no pricing, customers or case studies published	Google Play
Not to confuse with	Enerlytics of Muscat, Oman, a separate firm	enerlytics.om

Team

- Samir Ahmad, CEO and Co-founder, is almost certainly the "Sameer" in the meeting notes (name and role match; confirm). LinkedIn shows a Fulbright-funded sustainable-energy degree from UMass Lowell and earlier power-plant work (LinkedIn).
- Mujtaba Tasneem, Product Developer and Co-founder. His support email uses energyfuturesconsulting.com, so Enerlytics is probably a sister of Energy Futures Consulting (unverified).
- Talha Kamal (frontend), Ifra Mehmood (mobile) (About); Muhammad Taha, Senior Application Engineer (RocketReach).

Modules and capabilities

Module	What it tracks or does (as marketed)
BESS	SoC, pack voltage spread, cycles, availability, round-trip efficiency, levelised cost of storage, SoH, predictive maintenance

Module	What it tracks or does (as marketed)
Generator	Engine speed, real-time fuel use (kg/kWh), blended cost (PKR/kWh), active power, load ratio, power factor, rotor-health prediction
Solar PV	PR, curtailment, mismatch and soiling losses, cleaning alerts, O&M SLA tracking, remote inverter tests
Grid	Maximum demand, grid availability and load shedding, load-shedding forecast, net-metering reports, energy mix, power quality
Wind	Real-time and forecast generation
Utilities	Pump, motor, compressor, chiller, boiler; specific consumption, LCOE, production baseline
Platform	Predictive alerts; "AI and ML to support your decision making"; load and bill forecasting; carbon accounting; ISO 50001 and M&V support; "device agnostic" with no device list

Sources: Energy assets, Features, Home page.

What it does and does not do

It observes: dashboards, KPIs, alarms, forecasts of load, bills and load shedding, and rupee metrics such as blended PKR/kWh. Its "Microgrid Friendly" use case "monitors load profile and dispatch patterns to optimize generating assets", and it can run remote inverter tests.

No page describes TOU battery scheduling, a solver, or setpoints sent to inverters, batteries or gensets. No device list, edge gateway or API documentation is published.

Channels

- Novel Experience Est. holds "an authorized distributorship for Saudi Arabia" (page dated July 2024) (novelexpksa.com).
- A white-label portal, "AI Energy Asset Management Platform", sits at wateen.enerlyticslab.com, suggesting a Wateen Solar deployment. It is robots-blocked, so unverified (wateen.enerlyticslab.com).

Reading "observe and dispatch"

"Observe and dispatch" most likely means observing assets, then reporting on or recommending dispatch through alerts, KPIs and O&M actions. It does not appear to mean executing an optimised schedule. Product 1's increment is the closed loop: forecast, optimise against tariff, outage and fuel cost, send setpoints, verify the saving.

Enerlytics today vs Product 1 increment

Capability	Enerlytics today (evidence)	Product 1 increment
Data acquisition	Cloud platform, "device agnostic", no list	Uses Enerlytics or Product 2 data at 1-minute resolution
Battery state	Reports BMS SoC, SoH, cycles	Independent SoC check with confidence band and BMS trust score
Genset	Fuel rate, blended cost, load ratio	Fuel curve inside the optimiser; start, stop and loading decisions
Load shedding	Availability log and forecast	Reserve SoC sized to the next forecast outage
Forecasting	Load, bill, wind and PV forecasts	Probabilistic 15-minute PV, load and outage inputs to the optimiser
Tariff	Net-metering reports, MDI	TOU windows, ICP, net billing and MDI charge in the objective
Decision	Recommendations, "decision support"	MILP plan every 15 minutes over 48 hours
Control	Remote inverter tests only	Plan sent as inverter TOU slots or setpoints, with fallback
Proof	KPI reports, M&V use case	Saving vs the inverter's own mode at 1-minute resolution

Partner or competitor

Option	What each side brings	For	Against
A. Engine partner via API (recommended first)	Enerlytics: sites, data, dashboards; OnyxLeaf: dispatch engine and savings proof	Fastest route to pilot data	Depends on their data quality and API
B. White-label engine inside Enerlytics	As A, under their brand, like the apparent Wateen portal	Reaches their customers and Saudi distributor	Lower margin; OnyxLeaf brand hidden
C. Compete	OnyxLeaf builds monitoring plus dispatch	Owns the customer	Fights Enerlytics and NeoTAQ at once; rebuilds commodity monitoring

Test option A by asking Samir Ahmad for 2 to 3 sites with batteries and gensets, plus a data API. The risk is that Enerlytics adds optimisation itself, since it already markets AI and ML.

Who else is doing this

At least 7 Pakistani players and about 20 global ones monitor or control PV, battery and genset sites. None found publishes forecast-driven TOU optimisation across mixed brands in Pakistan; most dispatch is rule-based.

(a) Pakistan players

Player	What it does	Optimisation method	Brands	Customers and deployments	Pricing	PK / KSA	Link
NeoTAQ Systems (closest competitor)	neoCore edge gateway (RS485, Modbus, OPC-UA, 4G); neoSync solar-diesel; neoES battery dispatch; neoSphere cloud	Rules plus a learning "recommendation layer"; genset start/stop on SoC thresholds; edge AI and fleet forecasting planned	"OEM-agnostic", no list	Claims 50+ PK sites, 100 MW+; logos PepsiCo, GIKI, PSO, UNDP; ~30% cost cut at Fatima Rice (GFE claim)	Not published	HQ Masdar City; PK sites; Solar & Storage Live KSA, 23 to 25 Nov 2026	neotaqsys.com
Green Fuel Energy (channel)	Lahore EPC; sole PK distributor of NeoTAQ, distributor of Elum; solar-diesel and zero-export controllers	Rules: genset minimum 25 to 30%, solar first, "configurable priority rules and tariff schedules"	Multi-brand (Huawei, SMA)	GIKI 2 MWp, 20 inverters, 4 gensets; Haideri 1 MWp; Sharif Trust Hospital 1 MWp; 10 to 30% cost cut claimed	Not published	Pakistan	gf-energy.com
Enercon Systems (the "EnerCon")	Microgrid controls on Siemens PLC and PPC; genset sync; SCADA; cloud dashboard with AI load forecasts	Controls plus "energy mix optimization"; no TOU stated	Sinexcel, Siemens, Caterpillar	Al Karam (3 gensets, zero export); Sindh Feeds (4 × 233 kWh BESS); IPAK (5 gensets, 16 inverters)	Project-based	Karachi; Middle East and KSA listed	enerconsys.com
SkyElectric	Own inverter, battery, cloud, 24/7 NOC; SkyGrid EMS	SmartFlow AI uses tariff, load and "predicted grid availability"	Own only	10,000+ live systems	Not published	Pakistan; US office	skyelectric.com, ScyllaDB case
Reon Energy	SPARK telecom EMS; REFLEX BESS	Not stated; Lucky Cement BESS for PV smoothing, reserve, daytime genset switch-off	Own, CATL-based	5,500 towers; Lucky Cement 20.7 MW / 22.7 MWh	Energy-as-a-service	Karachi hub; HQ London, Dubai; Saudi SPARK licence	reonenergy.com, ESS News

Player	What it does	Optimisation method	Brands	Customers and deployments	Pricing	PK / KSA	Link
Huawei (with Jazz)	Telecom hybrid power; FusionSolar	"Intelligent energy management", closed	Huawei	1,000 Jazz sites, 13 MW; "as much as 96%" cost cut claimed	Bundled	PK; Saudi web page	Huawei
Nizam Energy	EPC, IPP, distributor	"AI software" for O&M faults	GoodWe, SAJ, TBB, Sunwoda	5,000+ projects	Not published	Pakistan	nizamenergy.com
Others	Hoymiles HiOne with built-in EMS; Feroze Power BESS; edotco "Green Energy" (2,000+ towers)	Not stated	Own	IBA, Tapal, PSO (Feroze)	Not published	Pakistan	Hoymiles, Feroze, edotco

"NeoSat Renewable" from the meeting notes matched no Pakistani renewable company; it is most likely NeoTAQ (confirm).

(b) Global C&I and residential optimisers

Player	What it does	Method	Brands	Deployments	Pricing	PK / KSA	Link
Tesla Opticaster	TOU, demand charges, export limits for Powerwall to Megapack	Statistics, ML, numerical optimisation; edge plus cloud	Tesla	"Over a hundred million hours"	Included	Saudi-locale pages only	Tesla
Schneider Microgrid Advisor	Forecasts, tariffs, demand charges, PV, battery and genset control	Forecast scheduling; 48-hour local fallback	Modbus, BACnet devices	Lidl Finland, others	Not published	UAE page	Schneider
Stem PowerTrack	Moved to utility-scale EMS and SCADA	Not detailed	Multi-vendor	38.3 GW solar, 1.8 GWh storage; managed-services ARR flat	~$1.1/kW-yr implied	None	Nasdaq
GridBeyond	C&I battery and flexibility trading	Solvers and RL	Multi-vendor	5 GW, 1,400+ sites	Not published	None	GridBeyond
Huawei FusionSolar	AI scheduling, peak-valley arbitrage	Forecast plus exchange prices	Huawei	20,000+ homes; €160 over 8 months claimed	Not published	Saudi page, European examples only	Huawei

Player	What it does	Method	Brands	Deployments	Pricing	PK / KSA	Link
Sungrow, SolarEdge, Enphase	24-hour plans against exchange prices	"AI" forecasting	Own	Europe, Australia	Not published	SolarEdge JV in KSA (2023)	Sungrow, SolarEdge
Victron Dynamic ESS	Forecast and price scheduling via VRM cloud	Closed; users report poor results under fixed UK TOU	Victron	Every Victron ESS	Free	Not stated	Victron, Community
Amber, Evergen, Tibber	Retailer or aggregator home-battery optimisation	Forecast plus wholesale price	Tesla, Sungrow, Huawei, others	AU, NL, DE	Amber A$25/month; Evergen free	None	Amber, Evergen

(c) Off-grid, microgrid and telecom controllers

Player	What it does	Method	Brands	Deployments	Pricing	PK / KSA	Link
Elum ePowerControl	PV-diesel-battery-grid control, zero export, peak shaving	Rules: genset minimum ~30%, SoC and ramp limits	"95% of equipment"	PK 200 kWp chemical plant; South Africa, Malawi	Quote only	PK via GFE	Elum
ComAp InteliNeo, InteliLite	Microgrid and telecom genset control	Battery-status genset cycling	Third-party gensets via Modbus	Not named	Feature SKUs	Not stated	ComAp
Eltek Smartpack2 (Delta)	Telecom hybrid controller	Genset at 80 to 90% load, cyclic	Own rectifiers	"Thousands of installations"; up to 80% opex cut claimed	Not published	Not stated	Eltek
Huawei telecom hybrid	Solar, diesel, grid, lithium dispatch; anti-theft	Closed	Huawei	Jazz 1,000 sites	Bundled	PK	Huawei Carrier
Victron Cerbo GX	Off-grid control, genset auto-start	Rules on SoC, load, voltage, time	Victron plus listed third parties	Large base	Hardware	Not stated	Victron
Enerwhere / Enlite	Leased solar-diesel-battery hybrids	Proprietary controller	Own	DIFC: ~27% less diesel (2017)	Lease	UAE	pv magazine
NEC telecom EMS	Forecasts PV, outage duration and end; plans genset and lithium	Forecast-based server plan	Own	20 Indian test sites; no saving reported	Trial only	India	NEC 2016

Player	What it does	Method	Brands	Deployments	Pricing	PK / KSA	Link
SolarAssistant	Raspberry Pi monitor and settings	Monitoring, manual settings	16 brands incl. Deye, Growatt, Voltronic	Large community	USD 59 to 229 one-time	Does not ship to PK or KSA	SolarAssistant

Evidence of value over the inverter's own mode

Independent evidence says the optimiser's edge is small under plain TOU, and grows with demand charges, outages and genset costs.

- With tiered peak charges, full MPC saved 13.9% and a peak-shaving rule 5.2%; perfect foresight reached 15.4% (simulation) (Pérez-Piñeiro et al., 2023).
- With real day-ahead forecasts, 24-hour MPC fell to about manual-dispatch value; forecast error raised demand charges 52% (simulation) (Mirletz & Laws, NREL 2023).
- Re-scored at 1-minute resolution, MPC's edge over rules shrank 69% on average for hourly schedulers (simulation) (Pinter et al., 2025).
- On real hardware in 4 reproduced homes, rule-based, decision-tree and MPC controllers landed within 0.6% of each other (lab) (Ruddick et al., 2024).
- Genset-hybrid claims of 27% to 96% compare against diesel-only operation, not a simpler controller (vendor claims).

No vendor A/B against the same inverter's built-in TOU mode was found. No peer-reviewed field trial of predictive versus rule-based genset dispatch was found either.

Pricing benchmarks

Model	Example	Price signal
Free, bundled	Tesla, Victron, Evergen	Rs 0 anchor for homes
Subscription	Amber SmartShift	A$25/month vs A$397 to 1,020/yr claimed upside: fee is 30 to 75% of upside
One-time licence and kit	SolarAssistant	USD 59 to 229; paid updates abandoned
Hardware with monitoring	Smartfox Pro 2	EUR 1,130 incl. 10 years monitoring (Galaxus)
C&I software ARR	Stem	~$1.1/kW-yr monitoring; ~$11/kWh-yr managed storage (upper bound, author's arithmetic)

Model	Example	Price signal
Performance contract or lease	Smart Joules (India), Enerwhere (UAE)	Guaranteed savings or no upfront cost (Outlook Business)
Project quote, feature SKUs	Elum, ComAp	Quote only

No Pakistan or Saudi EMS subscription price was found. A per-site fee looks viable only where gensets, outages or MDI exposure exist, or as a share of verified savings.

Open-source codebases

About 40 repositories were reviewed on 30 and 31 October 2026, and two MIT MILP engines, EMHASS and evcc-io/optimizer, can seed Product 1's core. Only REopt.jl (Julia, Apache-2.0) and python-microgrid (LGPL-3.0) model a fuel-costed genset with grid outages.

Comparison table

Stars are from GitHub pages on 30 to 31 October 2026; last commits are from git clones on 30 October 2026. "Not found" means the review found no such feature; "–" means not applicable.

Repo	Licence	Stars	Last commit	Language	Method / solver	Step and horizon	Genset	Outages	Degradation	Reuse verdict
davidusb-geek/emhass	MIT	668	2026-10-30	Python	MILP, CVXPY, HiGHS	30 min default; day-ahead and "naive MPC"	No	Not found	PWL stress cost	Base candidate
evcc-io/optimizer	MIT	90	2026-10-28	Python	MILP, PuLP, CBC or HiGHS; soft constraints	15 min, 2 days	No	Not found	No	Base candidate
NatLabRockies/REopt.jl	Apache-2.0	48	2026-10-28	Julia/JuMP	MILP, HiGHS; MPC module	time_steps_per_hour	Yes: fuel curve, on/off, turndown	Yes: start, duration, probability	SoH in MILP	Formulation reference
OpenEMS/openems	Edge EPL-2.0; UI AGPL-3.0	1.6k	2026-10-30	Java/OSGi	Genetic algorithm (Jenetics)	15 min, 24 h rolling	CHP-by-SoC only	Partial: reserve, off-grid switch	No	Reference, licence care
Akkudoktor-EOS/EOS	Apache-2.0	1.7k	2026-10-30	Python	Genetic algorithm (DEAP)	60 or 15 min; 24 h + 48 h tail	No	Not found	LCOS per kWh	Reference (providers)
springfall2008/batpred (Predbat)	Personal, non-commercial, UK only	329	2026-10-30	Python + C++	Heuristic search plus simulation	5-min simulation	No	Not found	Cycle cost	Avoid as code

Repo	Licence	Stars	Last commit	Language	Method / solver	Step and horizon	Genset	Outages	Degradation	Reuse verdict
victronenergy/dbus-systemcalc-py	MIT	17	2026-10-29	Python	None (DESS runs in cloud)	–	Genset delegate	Yes: load-shedding schedules	Battery-life delegate	Safety-layer reference
victronenergy/dbus_generator	MIT	2	2026-05-27	Python	Rules	–	Yes (rules)	–	No	Fallback reference
ahalev/python-microgrid	LGPL-3.0	115	2025-01-17	Python	CVXPY MILP MPC, rules, priority list	Horizon parameter	Yes: start-up, wind-down	Yes: grid status vector	Cycle cost	Benchmark only
FlexMeasures/flexmeasures	Apache-2.0	209	2026-10-30	Python	LP/MILP, Pyomo, HiGHS	Configurable	No	Not found	No	Architecture reference
corneel27/day-ahead	Apache-2.0	178	2026-10-27	Python	MILP, python-mip, CBC; XGBoost PV	Hourly	No	Not found	No	Reference
ADGEfficiency/energy-py-linear	GPL-3.0	103	2026-02-15	Python	MILP, PuLP, CBC	Interval data	CHP asset	Not found	No	Learn only
SESAM-Polimi/MicroGridsPy-SESAM	EUPL-1.2	31	2024-11-20	Python/ Pyomo	LP/MILP sizing	Sizing	Yes	Not found	No	Ideas only
CityLearn	MIT	655	2026-09-22	Python	RL environment	1 h default	Not found	Unverified (v2)	Capacity-loss coefficient	Benchmark
NatLabRockies/BLAST-Lite	BSD-3	79	2026-06-24	Python/ MATLAB	Empirical fade models	–	–	–	LFP 250 Ah model	Library
NatLabRockies/pysam	BSD-3	155	2026-09-27	Python/ C++	Battery dispatch and life model	Hourly, sub-hourly	–	–	Life model	Backtest library
pvlib/pvlib-python	BSD-3	1.7k	2026-10-28	Python	PV physics	–	–	–	–	Library
skforecast / darts / statsforecast	BSD-3 / Apache-2.0 / Apache-2.0	1.5k / 9.5k / 4.9k	Oct 2026	Python	Forecasting	–	–	–	–	Libraries
OpenSTEF/openstef	MPL-2.0	169	2026-10-29	Python	XGBoost, LightGBM pipeline	–	–	–	–	Library, unmodified
Quartz Solar Forecast	MIT	158	2026-09-06	Python	Boosted trees on NWP, UK-trained	Hourly, 15 min	–	–	–	Benchmark
kellerza/sunsynk / ha-solarman	Apache-2.0 / MIT	340 / 516	2026-10-24 / 2026-05-03	Python	Deye, Sunsynk, Solarman register maps	–	–	–	–	Executor reference
emoncms/emoncms	AGPL-3.0	1.3k	2026-10-21	PHP	Monitoring	–	–	–	–	Avoid

Per-repo reviews

- **EMHASS:** `src/emhass/optimization.py` (8,990 lines) is a CVXPY MILP with demand charges, inverter curves, a PWL `battery_stress_cost` and `perform_naive_mpc_optim`. It lacks a genset, outages and a 15-minute default, and is tied to Home Assistant, so lift patterns rather than fork.
- **evcc-io/optimizer:** `src/optimizer/optimizer.py` (about 1,446 lines) is a 15-minute PuLP MILP that "never returns infeasible" because limits are soft penalties. It runs as a Flask service with 19 test files: the best seed for a microservice engine.
- **REopt.jl:** `src/constraints/generator_constraints.jl` gives fuel slope, on/off intercept and minimum turndown; `src/core/electric_utility.jl` models outages. `src/constraints/battery_degradation.jl` tracks SoH and `src/mpc/model.jl` holds an MPC module; port these patterns to Python.
- **OpenEMS:** the TOU controller (`io.openems.edge.controller.ess.timeofusetariff`) picks BALANCING, DELAY_DISCHARGE, CHARGE_GRID or DISCHARGE_GRID per period via a genetic algorithm in `Simulator.java`. Its drivers cover GoodWe, Huawei, SMA, Sungrow and Victron; licences limit it to a reference.
- **Akkudoktor EOS:** a broad Apache-2.0 `prediction/` folder of PV, load and price providers. Its genetic algorithm (`optimization/genetic/genetic.py`) suits auditable commercial dispatch poorly.
- **Victron delegates:** `delegates/loadshedding.py` handles 4 scheduled outage windows with `minsoc` and `preparetime`; `startstop.py` in dbus_generator stacks SoC, load, voltage and tank conditions. Both are MIT and map onto Pakistani load-shedding fallback logic.
- **python-microgrid:** `src/pymgrid/algos/mpc/mpc.py` is a CVXPY MPC with a boolean `u_genset`, grid-status vector and `fuel_cost`. It is LGPL-3.0 and quiet since January 2025, so use it unmodified in tests only.
- **Predbat (avoid):** its licence allows only "personal, non-commercial purposes within the United Kingdom", and `deye.py` admits "no model of a generator". Only its published ideas, PV10/PV90 weighting and a cycle cost, may inform design.
- **FlexMeasures:** Pyomo with HiGHS in `flexmeasures/data/models/planning/`, a multi-tenant API and 189 test files; an architecture reference for the cloud service.
- **Forecasting libraries:** pvlib supplies the physical chain; skforecast, darts and statsforecast supply GBDT and baseline models with quantiles. Quartz is a benchmark only, being UK-trained.
- **BLAST-Lite and PySAM:** BLAST-Lite ships `lfp_gr_250AhPrismatic_2019` but calls its models "qualitative" beyond test data. PySAM's `BatteryStateful` is a step-wise battery and life simulator for backtests.

Recommended composition

1. Forecasting: pvlib baseline plus a skforecast or darts GBDT residual model with quantiles; Quartz as a benchmark.
2. Optimiser: OnyxLeaf's own Python MILP on CVXPY or PuLP with HiGHS, 15-minute steps, 48-hour receding horizon, built from EMHASS and evcc-io/optimizer structure.
3. Genset: REopt-style fuel slope and intercept, on/off and minimum turndown, plus minimum up and down times written in-house.
4. Outages and wear: grid import forced to zero in outage windows; a BLAST-Lite-derived PWL wear cost.
5. Executor and safety: re-implement the Victron load-shedding and genset rules; take Deye register references from kellerza/sunsynk.
6. Benchmarks: python-microgrid (unmodified) and CityLearn.

Licence rules

1. MIT, BSD, Apache-2.0: copy and modify, keep notices; do not use the "REopt" trademark.
2. MPL-2.0 (OpenSTEF): use unmodified; publish any modified MPL files.
3. LGPL-3.0 (python-microgrid): dynamic linking in test tooling only.
4. EPL-2.0 (OpenEMS Edge): read for patterns; modified EPL files stay EPL.
5. GPL-3.0 and EUPL-1.2 (energy-py-linear, MicroGridsPy): learn only, copy nothing.
6. AGPL-3.0 (emoncms, OpenRemote, OpenEMS UI): never embed in a hosted product.
7. Predbat and evcc's 89 sponsor-only files: do not copy.
8. Check Solcast, Forecast.Solar and Open-Meteo commercial terms before production.

Methods

The proven design is a MILP economic MPC at 15-minute steps over a 36 to 48-hour horizon, with a rule layer underneath. RL trails MILP by 22% to 35% on cost in like-for-like simulation, so it is not a first-release method.

Evidence labels: SIM (simulation), LAB (test rig), FIELD (real sites), VENDOR (commercial claim). Items marked "not re-verified" came from researcher knowledge with DOIs; check before quoting.

Forecasting

PV: the standard chain is an irradiance forecast, a physical model (pvlib), then a per-site ML correction. On weather-model inputs, tuned kernel ridge and MLP models reached 44.6% skill over persistence at 16 Hungarian plants (Markovics & Mayer, 2022).

- Solcast reports day-ahead daily PV energy MAPE of 4.4% at arid tropical sites; +24-hour GHI nRMSE is 16.7% arid and 34.8% humid (VENDOR) (Solcast).
- Lahore soiling ran about 0.8%/day at 30° tilt, with weekly cleaning optimal (FIELD) (Ullah et al., 2020). Model a soiling state from clear-day output.
- Winter smog makes weather-model forecasts over-predict; satellite blends and local bias correction help (inference).
- Score with daytime nRMSE or skill vs smart persistence, not MAPE (Yang et al., 2020, not re-verified).

Load: gradient-boosted trees on calendar, lag and weather features remain the workhorse (Miller et al., 2020, not re-verified). Transformers pretrained on 900,000 simulated buildings forecast real commercial buildings well zero-shot, a cold-start route (BuildingsBench, 2023). Load history is censored during outages, so training data must flag them (inference).

Outages: take the DISCO feeder schedule as input, then learn a per-feeder probability of unscheduled outages. NEC's Indian tower EMS forecast outage duration and end time to plan genset and battery (FIELD demonstration, no saving reported) (NEC, 2016). No study modelling Pakistani load-shedding schedules inside an MPC was found.

Optimisation

Method	Role in Product 1	Evidence
Rule baseline	Inverter's own self-use or TOU mode; HOMER load-following and cycle-charging at genset sites	Cycle charging runs gensets near rated output and charges to a set-point SoC (HOMER)
MILP day-ahead	Daily plan; also the "TOU slot" output for MVP mode	Perfect forecasts overstate value 1% to 8%; peak shaving is most robust to error (SIM) (Mirletz & Laws, 2023)
MPC rolling horizon	Main engine: 15 min, 36 to 48 h, re-solved every 5 to 15 min	Horizons of 24 h or less discharged too early for demand charges; 48 h preferred (SIM, 98 sites) (Neubauer & Simpson, 2015)
Two-layer MPC	15-minute MILP above a 1-minute rule layer	27.8% lower annual cost than single-layer MPC (SIM; one-day LAB) (Elkazaz et al.)

Method	Role in Product 1	Evidence
Stochastic MPC	Version 2: scenarios or quantiles for PV and outages	Beat deterministic MPC under uncertainty; no percentage given (SIM) (Blonsky, NREL 2020)
Reinforcement learning	Not in v1; later as an MPC imitator on solver-less devices	PPO, SAC, DDPG cost 22.5% to 35% above MIQP and broke power balance (SIM) (Shengren et al., 2022); CityLearn 2022 winners found optimisation generalised better (Nweye et al., 2023)

Genset-aware MPC for islanded sites follows Parisio et al. 2014 (LAB) and Sachs & Sawodny 2016 (SIM), both not re-verified.

Genset and outage modelling

- Fuel: HOMER's linear curve, fuel per hour = intercept × rated kW × on + slope × output kW. The worked example uses 0.033 and 0.273 L/h per kW (HOMER docs).
- Unit commitment: on/off binary, minimum load (25% to 30% in GFE and Elum controllers), start cost, minimum up and down times (Carrión & Arroyo, 2006, not re-verified).
- Run-hour maintenance cost and a daily run-hour cap.
- Outages: grid availability is a 0 or 1 input per step that bounds import. A reserve SoC covers critical load through the next forecast outage; scenarios generalise this (Khodaei, 2014, not re-verified).

Battery state and degradation cost

The optimiser must treat BMS SoC as a noisy measurement. On the LFP plateau, 1 mV of voltage error can mean about 5% SoC error (LAB) (Paizulamu et al., 2024). One user's JK BMS read 85% on an empty pack (FIELD anecdote) (Victron Community).

- Start with a throughput cost: replacement cost over lifetime throughput, about Rs 7.5 to 10.8/kWh at 6,000 cycles (assumption).
- Upgrade to a piecewise-linear cycle-depth cost, which nears rainflow accuracy with enough segments (SIM) (Xu et al., 2018).
- Add a calendar penalty on high SoC at high temperature, given 40 to 50 °C summers (inference).
- Calibrate per site from BLAST-Lite and rainflow on real SoC traces; no field study proving life gains from wear-aware dispatch was found.

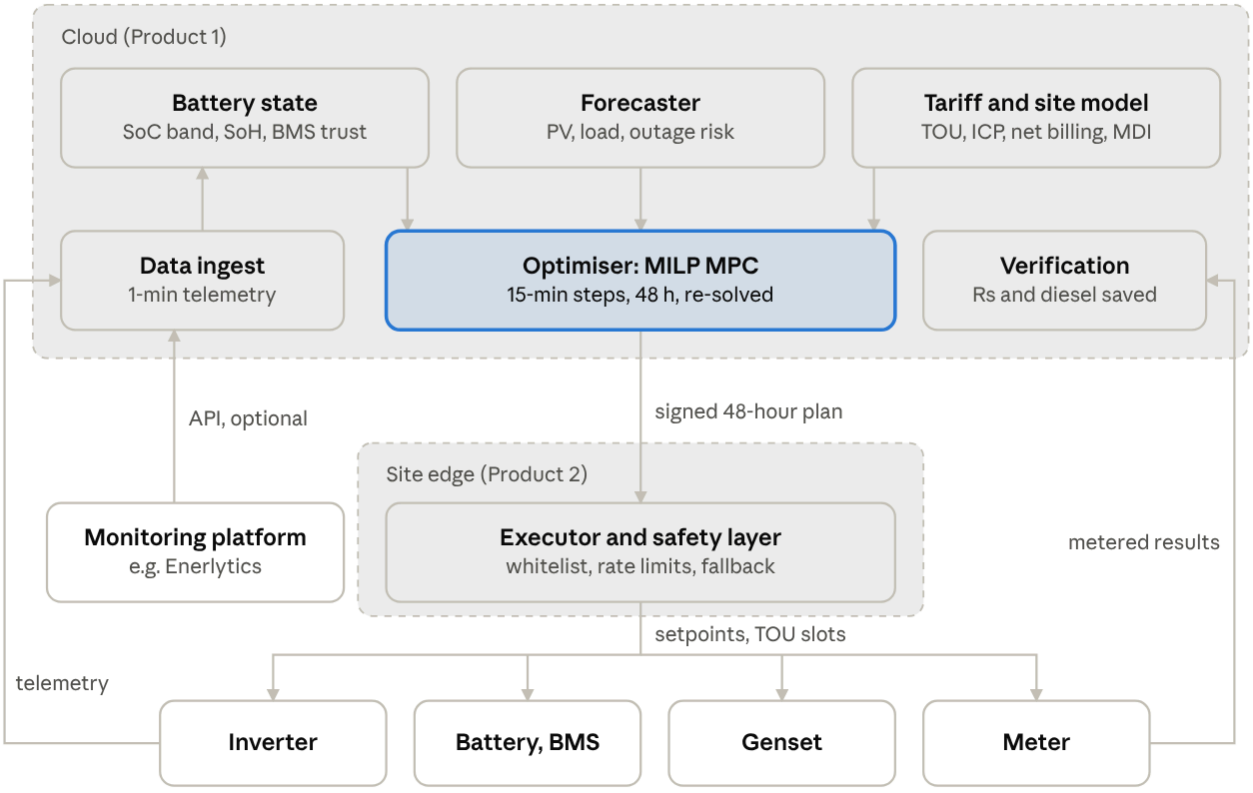
Formulation sketch

Element	Content
Time	15-minute steps, 48-hour horizon (192 steps), re-solved every 15 minutes
Decision variables	Grid import; grid export; battery charge and discharge; SoC; genset on, start and output; PV curtailment; unserved critical load (slack); month's peak import
Objective: minimise	Import × TOU or flat price (ICP price above baseline) − export × credit + genset fuel × diesel price + start and run-hour cost + battery wear + MDI charge on any rise above peak-to-date + penalty × unserved load − terminal value of stored energy
Power balance	PV − curtailment + import + discharge + genset = load + charge + export
Grid	Import ⩽ availability × limit; export ⩽ cap (zero-export option); no simultaneous import and export
Battery	SoC update with efficiencies; SoC within limits; power within inverter and BMS limits
Reserve	SoC ⩾ critical energy for the next forecast outage, divided by efficiency
Genset	Minimum load × on ⩽ output ⩽ rated × on; start ⩾ on now − on before; minimum up and down times; run-hour cap; "may charge battery" flag
Solver	HiGHS via CVXPY or PuLP; soft penalties so the model never returns "infeasible"

Proposed scope and architecture

The MVP has 7 components and two delivery modes. Mode A writes a daily schedule into the inverter's own TOU slots; Mode B sends 15-minute setpoints through Product 2.

The optimiser plans in the cloud; the edge executes and falls back



Product 1 architecture · 7 components, cloud and edge

The optimiser is the only component that decides; every other box feeds it or carries out its plan, and the edge reverts to the inverter's own mode when a plan expires.

Data flow: every 15 minutes the cloud optimiser pulls state, forecasts and tariff vectors, solves, and pushes a signed 48-hour plan to the edge. The edge executor follows the current step, corrects residuals with a 1-minute rule layer, and reverts to the inverter's own mode when the plan expires. Schneider's Microgrid Advisor keeps a similar 48-hour local fallback (Schneider).

Mode A suits the pilot. Deye hybrids expose 6 programmable time slots with SoC targets, and Huawei LUNA2000 exposes TOU period blocks (kellerza/sunsynk; huawei-solar-lib). A few writes a day avoid the flash wear SMA warns of for cyclically changed parameters (SMA Modbus TI).

#	Component	Inputs	Outputs	Build from
1	Site and tariff model	Tariff category, seasonal TOU windows, ICP baseline, billing regime and expiry, MDI basis, export cap, genset curve, critical load	Price and limit vectors per 15 min	Own code; REopt `electric_tariff.jl` pattern
2	Data ingest and store	1-minute telemetry from Product 2, Enerlytics API or OEM clouds; outage events; fuel logs	Clean series with outage flags	Own store, not emoncms
3	Battery state service	Pack voltage, current, temperature, BMS registers	SoC with confidence, usable kWh, SoH, BMS trust score, wear-cost curve	EKF on equivalent circuit; BLAST-Lite
4	Forecaster	Weather (Solcast or Open-Meteo), history, DISCO schedule	PV and load quantiles; outage probability per step	pvlib, skforecast or darts
5	Optimiser	Outputs of 1, 3 and 4	48-hour plan: battery, grid, genset, reserve	CVXPY or PuLP with HiGHS
6	Executor and safety layer	Plan; device capability list	Mode, power limits, export cap, TOU slots with SoC targets, genset start request; read-back; fallback	Command set defined by P1, run on Product 2 edge

#	Component	Inputs	Outputs	Build from
7	Verification and reporting	Telemetry, baseline replay, tariff	Daily and monthly rupee saving, diesel litres, peak kW, KPIs	Own code; PySAM for replay

Interfaces:

- Product 2 to Product 1: normalised 1-minute telemetry and a per-device list of writable primitives (work mode, power limits, export limit, TOU slots).
- Product 1 to Product 2: a 48-hour schedule of abstract setpoints with validity time and fallback rules. Product 2 enforces write whitelists, rate limits and revert timers.
- Enerlytics-type monitoring to Product 1: telemetry and its load-shedding forecast; Product 1 returns plans and verified savings for display.

Methodology and validation

Each pilot site needs 4 weeks of 1-minute data before the optimiser runs, then at least 8 weeks of switchback against the inverter's own mode. Savings are scored at 1-minute resolution, because re-scoring at 1 minute shrank hourly schedulers' MPC edge by 69% in one study (Pinter et al., 2025).

Data requirements per site

- Last 12 bills, tariff category, sanctioned load, MDI history, billing regime and contract expiry.
- 1-minute PV, load, grid import and export, battery power, voltage, current, temperature, BMS SoC and SoH.
- Genset status, output, fuel log or tank level, and rating.
- Outage log with start and end times, plus the DISCO feeder schedule.
- Battery chemistry, usable capacity and BMS model; inverter models, firmware and writable registers.
- PV tilt, azimuth, kWp and cleaning dates; the critical-load list and expected backup hours.

Baseline

The baseline is the inverter's own mode as the installer set it, usually self-use with a fixed TOU or backup reserve. At genset sites, also test a tuned rule: HOMER-style cycle charging with a set-point SoC. Claims must beat the tuned rule, not just the default.

Pilot design

Phase	Duration	What happens
0. Observe	4 weeks	Inverter's own mode; collect data; validate the baseline replay
1. Shadow	2 weeks	Optimiser plans but does not act; compare with what the inverter did
2. Switchback	8 weeks minimum	Days randomly assigned to baseline or optimiser; both reset to a common SoC at 06:00 to limit carry-over
3. Full control	4 weeks	Optimiser daily; check stability and fallback

A second estimate comes from counterfactual replay: for each day, simulate the other policy on the realised PV, load and outages. The two estimates must agree within a tolerance set in WP7. Run at least one pilot across an outage-heavy month.

KPIs

KPI	Definition
Rupee saving	Reconstructed bill plus fuel plus wear, optimiser vs baseline; Rs/day and % of energy cost
Diesel	Litres, run hours, starts per day; average genset load
Peak	Monthly MDI in kW
Forecast quality	Daytime PV nRMSE and skill vs smart persistence; load nRMSE; outage Brier score
Constraint violations	SoC below reserve at outage start; unserved critical load; genset below minimum load; export above cap
Battery use	Throughput kWh, equivalent full cycles, hours above 90% SoC in heat

KPI	Definition
Control health	Writes per register per day, read-back failures, fallback events
BMS trust	Gap between BMS SoC and independent estimate

Computing savings fairly

- Rebuild the bill at 1-minute resolution with the same tariff, TOU window, MDI rule and adders for both arms.
- Price diesel at measured litres where possible, else the fitted fuel curve, at the dated pump price.
- Charge battery wear to both arms at the same Rs/kWh.
- Report realised-forecast results and a perfect-foresight upper bound separately; state the GST treatment.

Success and kill criteria (proposed, to agree with the engineering manager)

- Success: at 2 of 3 sites, the verified increment over the tuned baseline is at least 3 times the planned monthly fee. The ratio follows Amber's fee of 30% to 75% of claimed upside (Amber).
- Success also needs zero unserved critical load caused by the optimiser and zero writes outside the whitelist.
- Kill if the increment is below the fee at all sites after 8 weeks of switchback.
- Kill or reshape if the two most common pilot brands refuse TOU-slot or setpoint writes.
- Reshape toward genset-only sites if grid-tied increments fall below 2% of the energy bill while genset sites clear the bar.

Risks and open questions

Thirteen risks matter most; the top one is that the measured increment over the inverter's own mode is too small to charge for.

Risk	Why it matters	Mitigation
Increment too small	Hardware test found controllers within 0.6%; self-use already captures most PV shift	Target outage and genset sites; offline backtest gate (WP7) before field control

Risk	Why it matters	Mitigation
Tariff and policy churn	Uniform schedule ends December 2026; ICP under review; net-billing amendment unconfirmed	Tariffs as versioned data; re-run the value model quarterly
Wrong battery SoC	LFP plateau and cheap BMS sensing give large errors; reserve may be empty at outage start	Independent shunt where possible; SoC confidence band; wider reserve
Flash wear and inverter damage	SMA warns cyclic parameter changes destroy flash	Mode A daily TOU slots; volatile or dispatch registers; write budget; read-back
Unreliable outage schedules	Faults and fuel shocks are unscheduled	Learn outage probability from telemetry; reserve margin; stochastic MPC later
Forecast error	NREL found MPC fell to manual-dispatch value with real forecasts	Robust, peak-shaving-style plans; quantile forecasts
NeoTAQ adds forecasting	It plans edge AI and fleet forecasting, with a channel and 50+ sites	Move fast; consider partnering via GFE
Enerlytics builds or declines	Removes the data and channel route	Keep an own data path via Product 2
Free OEM features	Huawei, Victron and Tesla bundle scheduling free	Sell to mixed-brand C&I with gensets; sell proven savings
Warranty and liability	Third-party control may conflict with OEM terms	Installer terms; certified-control brand list; automatic fallback
Cybersecurity	SunSpec requires IP-based Modbus to be secured	Outbound-only links; signed plans; no grid-protection writes
Missing genset data	Many sites lack fuel sensors	Fuel logs and a CT on genset output; fitted fuel curve
Licence contamination	Predbat, GPL and AGPL code within reach	Licence rules above; code review gate

Open questions for the engineering manager

1. Is "NeoSat Renewable" in the notes actually NeoTAQ Systems?
2. Is Enerlytics a partner, a channel or a competitor, and will Samir Ahmad share a data API and 2 to 3 sites?
3. Which sites can Green Fuel Energy, Enercon or Enerlytics offer for pilots, with batteries and gensets?
4. The notes stress off-grid and remote sites; the research favours grid-tied C&I with gensets first. Which comes first?
5. When will Product 2 write to the first two brands, and may the pilot use inverter TOU slots or OEM cloud APIs before then?
6. Which fee model to test: per-site monthly, share of verified savings, or bundled through the installer?
7. Who carries liability for a control-caused outage: OnyxLeaf, the installer or the owner?
8. Does the independent SoC and SoH check belong to Product 1 or Product 2?
9. For B-2 and B-3 sites, does MDI set the fixed charge, and over what interval? A real bill will answer this.
10. Are pilot sites on net billing or legacy net metering, and when do contracts expire?
11. For Saudi Arabia, which route first: Novel Experience (Enerlytics' distributor), or IMSIU and MISA?
12. What build capacity exists, and who owns the pilot?

Work packages

Twelve work packages take Product 1 from tariff model to a verified field pilot in about 34 person-weeks (pw). That is roughly 6 calendar months with 2 to 3 engineers; effort figures are rough planning estimates, not sourced.

ID	Name	Goal	Inputs	Deliverable	Acceptance criterion	Depends on	Effort
WP1	Tariff and regime model	Encode tariffs, TOU windows, ICP, net billing, net metering, MDI; KSA flat tariffs	SRO 279(I)/2026, NEPRA SROs, pilot bills	Versioned tariff library and bill calculator	Reproduces each pilot site's last 3 bills within 2%	WP2 (bills)	2 pw

ID	Name	Goal	Inputs	Deliverable	Acceptance criterion	Depends on	Effort
WP2	Pilot sites and data audit	Sign 3 to 5 sites with PV, battery and preferably genset	Introductions via GFE, Enercon, Enerlytics	Site agreements; audit of brands, registers, data gaps	3+ sites signed, covering 2+ inverter brands and 1+ genset site	None	3 pw
WP3	Data pipeline and battery-state check	Clean 1-minute data and an independent SoC estimate	Product 2, OEM or Enerlytics feeds	Time-series store with outage flags; SoC service with trust score	14 days from 3 sites with under 2% gaps; BMS-vs-estimate gap reported	WP2	3 pw
WP4	Baseline replay and 1-minute bill engine	Simulate the inverter's own mode and tuned rules	WP1, WP3 data, inverter settings	Replay simulator (PySAM-based)	Daily grid import within 3% and battery throughput within 5% of measured on 90% of days	WP1, WP3	3 pw
WP5	Forecasters v1	PV, load and outage forecasts with quantiles	WP3 data, weather API, DISCO schedules	Forecast service	PV daytime skill 25%+ vs smart persistence over 4 weeks; load beats seasonal persistence; outage model beats schedule-only	WP3	4 pw

ID	Name	Goal	Inputs	Deliverable	Acceptance criterion	Depends on	Effort
WP6	Optimiser v1	MILP MPC with genset, outages, MDI, ICP and wear	WP1, WP5, BLAST-Lite	Optimiser service with tests	192-step genset case solves in under 30 s; zero hard-constraint breaches over 1,000 simulated days; matches a REopt reference case within 1%	WP1, WP5	5 pw
WP7	Offline backtest and go/no-go	Estimate the increment before field control	WP4, WP5, WP6	Per-site value report, realised and perfect forecasts	Written go/no-go against kill criteria; replay and measured baselines agree	WP4, WP6	2 pw
WP8	Executor and safety layer	Turn plans into safe device commands	Product 2 interface, register maps	Mode A for 2 brands; Mode B spec; fallback logic	72-hour bench test: 100% read-back success; fallback within one control period on link loss; write budget kept	WP6, Product 2	4 pw
WP9	Verification and reporting	Monthly rupee and diesel report	WP4, telemetry	Report generator and KPI dashboard	Reproduces savings from raw data; reviewed by one site owner	WP4	2 pw
WP10	Field switchback pilot	Measure the increment in the field	WP7 go, WP8, WP9	Pilot dataset and results write-up	8+ weeks of switchback at 2 sites; zero safety incidents; KPIs reported	WP7, WP8, WP9	3 pw over 18 weeks

ID	Name	Goal	Inputs	Deliverable	Acceptance criterion	Depends on	Effort
WP11	Channel and pricing track	Settle Enerlytics, GFE and Enercon roles and the fee model	Meetings, WP7 results	Partner terms; willingness-to-pay notes	Written positions from Enerlytics and one installer; price feedback from 5+ site owners	WP2	2 pw
WP12	Saudi scoping	Find remote diesel-hybrid prospects; confirm KSA tariffs	Novel Experience, IMSIU, SERA draft	KSA brief and prospect list	5+ named KSA remote-site prospects; KSA tariffs in WP1	WP1	1 pw

Order: WP1 and WP2 start together; WP3 to WP6 follow; WP7 gates any field control; WP8 and WP9 run alongside WP7; WP10 starts only on a "go"; WP11 and WP12 run throughout.

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- Profit, peak relief load shedding; The Nation, end of load management; ProPakistani, economic load shedding; Dawn, Punjab outages

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- Elum microgrid controller, Elum SD/SD+; ComAp telecom; Eltek brochure; Huawei hybrid power; Victron Cerbo GX; Enerwhere DIFC; SolarAssistant shop; Smartfox; Smart Joules

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- Neubauer & Simpson, NREL 2015; Elkazaz et al.; Blonsky, NREL 2020; Shengren et al. 2022; Nweye et al. 2023; Parisio et al. 2014; Sachs & Sawodny 2016
- HOMER fuel curve; HOMER cycle charging; Carrión & Arroyo 2006; Khodaei 2014; NEC telecom EMS 2016
- Xu et al. 2018; Paizulamu et al. 2024; Victron Community, JK BMS SoC; SMA Modbus TI

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- dbus-systemcalc-py; dbus_generator; python-microgrid; FlexMeasures; day-ahead; energy-py-linear; MicroGridsPy; CityLearn
- BLAST-Lite; PySAM; pvlib; skforecast; darts; statsforecast; OpenSTEF; Quartz Solar Forecast

- kellerza/sunsynk; ha-solarman; huawei-solar-lib; emoncms